

## 1. Unix Basics

### Unix Overview

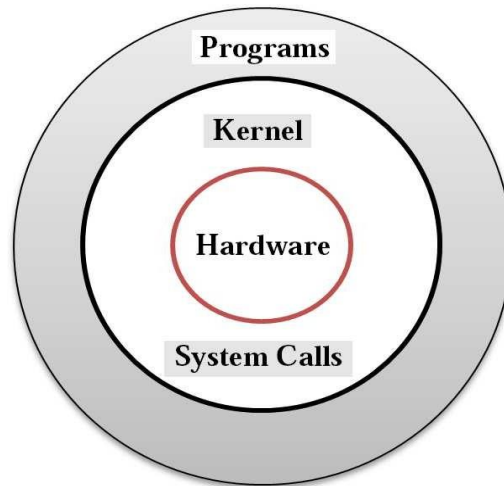
- **UNIX** is an innovative or groundbreaking operating system which was developed in the 1970s by Ken Thompson, Dennis Ritchie, and many others at AT&T Laboratories
- The foundation for many modern OSES like macOS, Linux, and Android. Key features include its hierarchical file system, a command-line shell for user interaction, and the "Unix philosophy" of small, interconnected tools used with pipes to perform complex tasks.
- It is like a backbone for many modern operating systems like Ubuntu, Solaris, Kali Linux, Arch Linux, and also POSIX. Originally, It was designed for developers only, UNIX played a most important role in the development and creation of the software and computing environments.
- The Unix operating system is a set of programs that act as a link between the computer and the user.
- The computer programs that allocate the system resources and coordinate all the details of the computer's internals is called the operating system or the **kernel**.
- Users communicate with the kernel through a program known as the **shell**. The shell is a command line interpreter it translates commands entered by the user and converts them into a language that is understood by the kernel.
- A user can also run multiple programs at the same time hence Unix is a multitasking environment.

### **Unix Introduction**

- Unix is multi user and multi-tasking operating system.
- All the commands in Unix should be given in lower case.
- Unlike Unix.
- Processor will be in protected mode in Unix.
- Unix which uses all the available RAM
- Unix needs an administrator.
- Unix employs time sharing operating system.
- Unix supports both character user interface and graphical interface(X Window)
- User requires legal username and password to use Unix machines.
- Unix used single directory tree (/) irrespective of how many drives or partitions are there.
- Till recently
  
- Device management using device driver
- Process management using process and threads
- Memory management
- File Management

In addition, all operating systems come with a set of standard utilities. The utilities allow common tasks to be performed as

- Being able to start and stop processes
- Organize files into sets as directories
- View files and sets of files
- Edit files
- Communicate between processes



**Figure 1.2** Linux Architecture

## **Features of Linux**

Here are some of the benefits and feature that Linux provides over single-user operating systems and other version of UNIX for the PC.

### ❖ **Multiuser System:** -

- Multiple users can log in and work on the same system simultaneously.
- Unix provides multiple programs to run and compete for the attention of the CPU. This happens in 2 ways. (1) Multiple users running multiple jobs (2) Single user running multiple jobs.
- In UNIX, resources are actually shared between all the users, so-called a multi-user system.

### ❖ **Multitask System:** -

- A single user may run multiple tasks concurrently. Example: Editing a file, printing another on the printer & sending email to a person, and browsing the net too at the same time. The Kernel is designed to handle user's multiple needs.
- Can run many programs/processes at the same time without interfering with each other.

## ❖ **Programming facility:** -

- Unix provides shell which is also a programming language designed for programmers, not for casual end-users. It has all the control structures, loops, and variables required for programming purposes. These features are used to design the shell scripts (programs that can invoke the UNIX commands).
- Provides compilers, editors, and debugging tools for application development.

## ❖ **Portable:** -

- Unix is highly portable, meaning it can be easily adapted to run on different hardware architectures.
- Written in **C language**, so it can run on different hardware with little modification.

## ❖ **Security:** -

- Unix has robust security features built-in, including user-level permissions and access control lists.
- Provides strong user authentication (username & password).
- File permissions (read, write, execute) to protect data.

## ❖ **TCP/IP networking support:**

- TCP/IP ("Transmission Control Protocol/Internet Protocol") is the set of protocols which links millions of computers into a worldwide network known as the Internet. With an Ethernet connection, one can have access to the internet or to a local area network from your Linux system. Or, using SLIP ("Serial Line Internet Protocol"), you can access the Internet over the phone lines with a modem.

## ❖ **Virtual memory and shared libraries:**

- Linux can use a portion of the hard drive as virtual memory, expanding the total amount of available RAM. Linux also implements shared libraries, allowing which use standard subroutines to find the code for these subroutines in the libraries at runtime. This saves a large amount of space, as each application doesn't store its own copy of these common routines.

## ❖ **Hierarchical File System:** - Organizes files in a tree-like directory structure, starting from the **root (/)**.

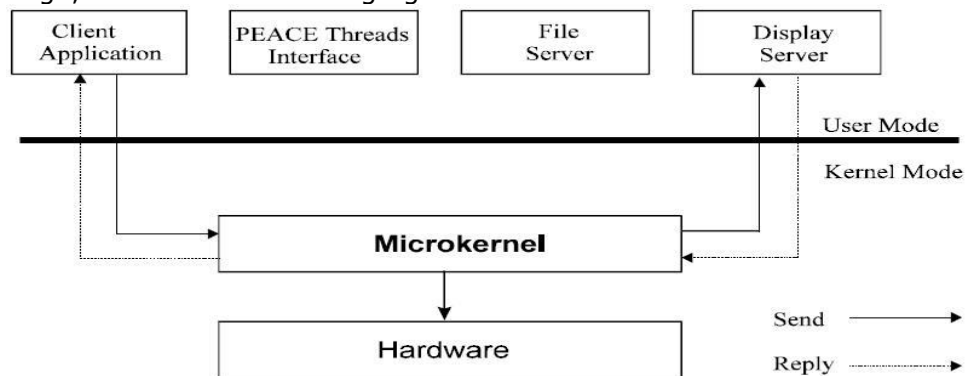
## **Kernel:** -

- A kernel is a central component of an operating system.
- It acts as an interface between the user applications and the hardware.
- The sole aim of the kernel is to manage the communication between the software (user level applications) and the hardware (CPU, disk memory etc).
- The kernel is a program that constitutes the central core of a computer operating system. It has complete control over everything that occurs in the system.
- A kernel can be contrasted with a shell (such as bash, csh or ksh in Unix-like operating systems), which is the outermost part of an operating system and a program that interacts with user commands.
- The kernel itself does not interact directly with the user, but rather interacts with the shell and other programs as well as with the hardware devices on the system, including the processor (also called the central processing unit or CPU), memory and disk drives.
- **The main tasks of the kernel are :**
  1. Process management
  2. Device management
  3. Memory management
  4. Interrupt handling
  5. I/O communication
  6. File system...etc...
- **Kernels may be classified mainly in two categories.**
  1. Monolithic
  2. Microkernel

- **Monolithic Kernels:** - Monolithic kernels, which have traditionally been used by Unix-like operating systems, contain all the operating system core functions and the device drivers (small programs that allow the operating system to interact with hardware devices, such as disk drives, video cards and printers).
- **Microkernel:** - A microkernel usually provides only minimal services, such as defining memory address space; inter process communication (IPC) and process management. All other functions, such as hardware management, are implemented as processes running independently of the kernel.

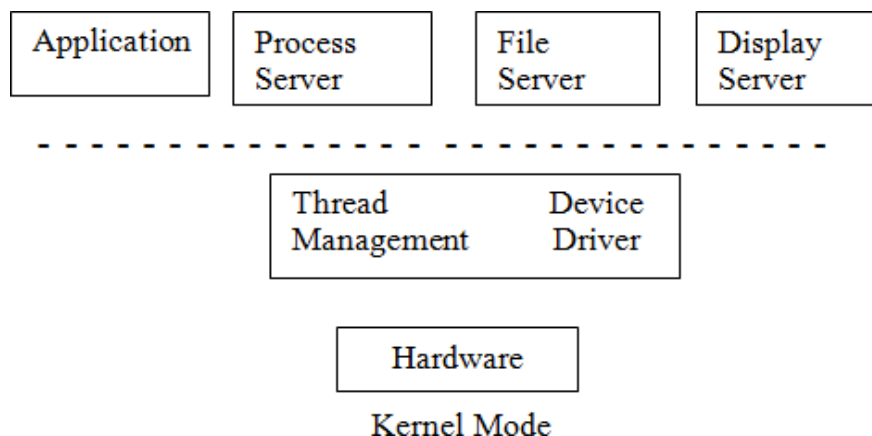
**Client-Server system (Microkernel system): -**

- The beginning of new concepts in operating system design, microkernel, is aimed at migrating traditional service of an operating system out of the monolithic kernel into the user-level process.
- The idea is to divide operating system into several processes, each of which implements a single set of services. For example, I/O servers, memory servers, process servers, threads interface system.
- Each server runs in a user mode, provides services to the requested client.
- The client, which can be either another operating system component or application program, requests a service by sending a message to the server.
- An OS kernel (microkernel) running in kernel mode delivers the message to the appropriate server; the server performs the operation; and microkernel delivers the result to the client in another message, as shown in following figure.



**Figure 1.3** Unix Kernel

Most of the currently prevalent Operating Systems use instead a *micro kernel*, of minimum size. Many traditional kernel operations are made into level services. Communication bearing service is often carried out by an explicit messagepassing mechanism. Mach is one of the major micro-kernel operating system whose concepts are used by many others (see Figure 1.4)



## Basic LINUX Commands

Following are the basic LINUX commands useful to day to day operations.

| No. | Command | Description                                                                                                                                                                 | Example                                                                          | Descriptions                                                                                                                                                                                                                                 |
|-----|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | cd      | Type cd followed by the name of a directory to access that directory. Keep in mind that you are always in a directory and allowed access to any directories hierarchically. | cd /var/log<br>cd ../cd<br>../root<br>cd Desktop cd<br>cd -                      | To change directory To go up a directory and change dir. Change directory To go to user's home directory To go to previous directory                                                                                                         |
| 2   | pwd     | To print current working directory on screen                                                                                                                                | pwd                                                                              | It will show full path of your current dir                                                                                                                                                                                                   |
| 3   | ls      | To list files in a directory                                                                                                                                                | ls -<br>lls -<br>Sl<br>ls -<br>ahl<br>ls<br>-ld ls<br>-t<br>ls -<br>tlrls<br>-il | Long list of files in a dir<br>Long list of files in a dir sorted by size<br>Long list of archive and hidden files in a directory To show only directory Sort by access/modify time<br>Time wise sorting of dir<br>Inode wise sorting of dir |
| 4   | cat     | To read file                                                                                                                                                                | cat filename<br>cat -n filename                                                  | Read file with all line nos.                                                                                                                                                                                                                 |
| 5   | more    | To read long file with Pagination                                                                                                                                           | more filename<br>cat filename   more                                             |                                                                                                                                                                                                                                              |
| 6   | cp      | Copy a file                                                                                                                                                                 | cp<br>cp -<br>Icp<br>-r<br>cp -r<br>-Pcp -<br>a<br>cp -av                        | Copy<br>Copy interactively<br>Copy recursively<br>Copy recursively & Preserve the permission<br>Copy all files with retaining permission<br>Copy all files & directories recursively with retaining permission                               |
| 7   | mv      | Move / rename file                                                                                                                                                          | mv file1 file2<br>mv file1 /home/user1/                                          | Rename file1 with file2<br>Move file1 to /home/user1/                                                                                                                                                                                        |

|    |       |                                                                                                                                                                                                                                                            |                                                                                                                                                             |                                                                                                                                                                                                               |
|----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8  | rm    | Remove files                                                                                                                                                                                                                                               | rm -<br>i rm<br>-r<br>rm -<br>f rm<br>-rf                                                                                                                   | Remove interactively<br>Remove recursively<br>Remove forcefully<br>Remove recursively & force fully                                                                                                           |
| 9  | mkdir | To create an empty directory                                                                                                                                                                                                                               | mkdir -p<br>/var/www/html/example<br>mkdir -p<br>/tmp/1/2/3/4/5/6/7/8<br>mkdir -p<br>user/{Jan, Feb, March}/{1, 2, 3, 4, 5}                                 | Create dir with permission of current user<br>To create multiple dir inside a dir<br>To create dir inside dir                                                                                                 |
| 10 | rmdir | To remove an empty Dir                                                                                                                                                                                                                                     | rmdir dirname                                                                                                                                               |                                                                                                                                                                                                               |
| 11 | grep  | Looks for pattern in file                                                                                                                                                                                                                                  | grep root<br>/etc/passwd grep -v<br>root /etc/passwd cat<br>/etc/passwd   grep<br>root grep -lr<br>password /etc/ grep<br>-rn password /etc/ or<br>filename | Show all matches of root<br>Show all lines that do not match root<br>Show all matches of root<br>show all files in /etc which contains word password<br>show line no & word password in /etc/ dir or filename |
| 12 | wc    | Word count                                                                                                                                                                                                                                                 | wc -l filename                                                                                                                                              | Shows no of lines in filename                                                                                                                                                                                 |
| 13 | Man   | To view the manual page for a command, type man                                                                                                                                                                                                            | man <command_name><br>man ls                                                                                                                                | Search for man pages whose descriptions contain a specific keyword.                                                                                                                                           |
| 14 | Chmod | The chmod command in Unix-like operating systems is used to change the access permissions of files and directories. The name "chmod" is an abbreviation of "change mode." These permissions determine who can read, write, or execute a file or directory. | chmod<br>[who][operator][permissions] file(s)                                                                                                               |                                                                                                                                                                                                               |
| 15 | Diff  | The diff command in Unix-like operating                                                                                                                                                                                                                    | diff file1.txt file2.txt                                                                                                                                    | The diff command's output indicates how to transform file1.txt to                                                                                                                                             |

|    |           |                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                |                                                                                                                                                                      |
|----|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |           | systems is a command-line utility used to compare two files line by line and display the differences between them. It is widely used for identifying changes in text files, generating patches, and managing code versions.                      |                                                                                                                                                                                                                | match file2.txt. The output uses specific symbols and instructions                                                                                                   |
| 16 | Wildcards | Unix-like operating systems, wildcards (also known as metacharacters) are special characters used in the shell to match multiple filenames or directory names based on a pattern. They are expanded by the shell before the command is executed. | <p>ls *.txt will list all files ending with .txt.</p> <p>rm report* will remove all files starting with report.</p> <p>cp * /backup will copy all files in the current directory to the /backup directory.</p> | These wildcards are commonly used with commands such as ls, cp, mv, rm, find, and many others, allowing for efficient manipulation of multiple files and directories |

## Process Management Commands

Following are the Process LINUX commands.

| No. | Command | Description                                                                                                         | Example                                                   | Descriptions                                                                                                                                                            |
|-----|---------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Ps      | This command stands for Process status of processes running for a user. Displays information about active processes |                                                           | Shows processes associated with the current terminal. Provides a comprehensive view of all running processes for all users, including those not attached to a terminal. |
| 2   | Kill    | The <b>kill</b> command in <b>Linux</b> is used to stop (terminate) processes by sending them                       | <p>kill &lt;PID&gt;</p> <p>kill [options] &lt;PID&gt;</p> | Sends the default TERM signal to the process with the specified Process ID (PID).                                                                                       |

|   |       |                                                                                                                                                                                                                                         |                                                                                                                  |                                                                                                                                                                                |
|---|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |       | signals. Despite its name, it doesn't always mean "kill"; it can send different signals to a process.                                                                                                                                   |                                                                                                                  |                                                                                                                                                                                |
| 3 | Nice  | the <b>nice</b> command is used to <b>start a process with a specific priority</b> .                                                                                                                                                    | nice<br>./my<br>prog<br>ram                                                                                      | Every process has a <b>priority</b> that the scheduler uses to decide which process runs first.<br><br>nice value ranges from -20 to 19, where -20 is of the highest priority. |
| 4 | Sleep | the <b>sleep command</b> is used to <b>pause (delay) execution</b> of a process or script for a specific amount of time. It doesn't run in the background by default—it simply suspends the process until the time elapses.             | sleep NUMBER[SUFFIX]<br><br>sleep 5<br><br>sleep 2m                                                              | The sleep command in Linux is used to pause the execution of a command or script for a specified period.                                                                       |
| 5 | Batch | the <b>batch command</b> is used to <b>schedule processes for execution in the background when the system load is low</b> . It's similar to <b>at</b> , but instead of running at a fixed time, it waits until the system is less busy. | batch<br># Enter commands, then<br>press Ctrl+D to submit the<br>job<br><br>batch <<br>/path/to/your/commands.sh | The batch command reads commands from standard input or a specified file and executes them only when the system's load average drops below a certain threshold                 |
| 6 | At    | the <b>at command</b> is used to <b>schedule one-time jobs</b> to run at a specific time in the future. It works with the <b>atd daemon</b> , which must be running on your system.                                                     | at [TIME]<br><br>at 9am tomorrow<br>at><br>/home/user/myscript.sh<br>at> <Ctrl+D>                                | At a <b>specific time</b> you set.                                                                                                                                             |